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(54) **METHOD FOR THE FERMENTATIVE PRODUCTION OF L-LYSINE USING C. GLUTAMICUM STRAINS EXPRESSING A HETEROLOGOUS NICOTINAMIDE NUCLEOTIDE TRANSHYDROGENASE PNTAB AND HAVING AN INCREASED ACTIVITY OF A MYO-INOSITOL PERMEASE**

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(57) **ABSTRACT**

A *C. glutamicum* strain has enhanced productivity of L-lysine, and a method produces L-lysine using the *C. glutamicum* strain. The *C. glutamicum* strain expresses NAD(P)(+) transhydrogenase subunit alpha PntA and NAD(P)(+) transhydrogenase subunit beta PntB combined with increased activity of the myo-inositol transporter IolT. A method produces L-lysine in a fermentative process employing such *C. glutamicum* strain.

Specification includes a Sequence Listing.

**METHOD FOR THE FERMENTATIVE
PRODUCTION OF L-LYSINE USING C.
GLUTAMICUM STRAINS EXPRESSING A
HETEROLOGOUS NICOTINAMIDE
NUCLEOTIDE TRANSHYDROGENASE
PNTAB AND HAVING AN INCREASED
ACTIVITY OF A MYO-INOSITOL
PERMEASE**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This patent application claims priority to European Patent Application No. 24161458.5, filed on Mar. 5, 2024, in the European Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

REFERENCE TO A SEQUENCE LISTING

[0002] The present application is accompanied by an XML file as a computer readable form containing the sequence listing entitled, "2024-12-19-SL-as-filed-006374US.xml", created on Dec. 19, 2024, with a file size of 72,224 bytes, the content of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

[0003] The present invention relates to a *Corynebacterium glutamicum* (*C. glutamicum*) strain which has enhanced productivity of L-lysine, and to a method for producing L-lysine using the same. In particular, the present invention concerns a *C. glutamicum* strain expressing NAD(P)(+) transhydrogenase subunit alpha PntA and NAD(P)(+) transhydrogenase subunit beta PntB combined with increased activity of the myo-inositol transporter IolT and a method for the fermentative production of L-lysine using such *C. glutamicum* strain.

[0004] Since L-Lysine remains a competitive market and has high commercial value, every percentage point increase in both production and productivity can bring huge benefits.

Description of Related Art

[0005] An important factor in the biotechnological production of L-lysine with microorganisms is the sufficient supply of NADPH. To produce one molecule of L-lysine, four molecules of NADPH are required to drive the necessary reducing steps. NADPH is provided from reactions of the pentose phosphate pathway (PPP) and the citric acid cycle (TCA).

[0006] To improve the availability of NADPH, production strains can be optimized by the expression of beneficial enzymes. For example, the introduction of an NADP-dependent glyceraldehyde 3-phosphate dehydrogenase (GapN) of *Streptococcus mutans* leads to increased NADPH availability in *Corynebacterium glutamicum* (*C. glutamicum*) and thus increased lysine yields (Takeno, S., Murata, R., Kobayashi, R., Mitsuhashi, S., & Ikeda, M. (2010) Applied and environmental microbiology, 76(21), 7154-7160). If an NADP dependent Gap enzyme is used instead of an NAD dependent one, energy conservation in form of ATP generation is omitted in this step, reducing the overall ATP yield per glucose.

[0007] Transhydrogenase activity offers an alternative perspective for sufficient NADPH supply, which is relevant for most amino acid production systems. One example is the use of the membrane-integral nicotinamide nucleotide transhydrogenase PntAB of *Escherichia coli* (*E. coli*), but also the use of PntAB of *Corynebacterium urealyticum* (*C. urealyticum*), which can use the electrochemical proton gradient across the cytoplasmic membrane to drive the reduction of NADP(+) via the oxidation of NADH. As *C. glutamicum* does not possess such an enzyme, the expression of the heterologous transhydrogenase PntAB gene results in an efficient way to improve the biotechnological production of metabolic compounds (Liu J, Li H, Zhao G, Caiyin Q, Qiao, J Ind Microbiol Biotechnol. 2018; 45(5):313-327). Since the native reactions providing NADPH (zwf, gnd, icd) are associated with carbon loss in the form of CO₂ evolution, and GapN has a higher strain on energy conservation, use of PntAB is a preferable route in terms of ATP, carbon yield and environmental impact.

[0008] There are PntAB variants from different organisms available. The PntAB variant from *E. coli* is well researched and characterized. Further examples of organisms harbouring pntAB variants are *Oenococcus oeni*, *Synechocystis* sp. and *Lactobacillus buchneri* (Liu S, Skory C, Liang X, Mills D, Qureshi N., J Ind Microbiol Biotechnol. 2019; 46(11): 1547-1556). The pntAB variant from *Corynebacterium urealyticum* (*C. urealyticum*), a closer relative to *C. glutamicum*, has to date not been investigated in such detail.

[0009] It can be shown that the heterologous expression of pntAB for example from *E. coli* or from *C. urealyticum* in *C. glutamicum* leads to increased lysine yields (cf. Experimental Section). However, without adjustment of sugar translocation into the cell, the specific productivity of the cells does not increase, which in turn leads to longer fermentation runs. The term productivity is a measure of how fast a product (i.e. L-lysine in the present case) is being produced and is calculated as the concentration of product produced over time in grams product per liter per hour (g/L/h).

[0010] However, the reduction of the required fermentation times will lead to reduced production costs and higher space time yields, offering significant economic benefits.

SUMMARY OF THE INVENTION

[0011] The object of the present invention is to increase the productivity with respect to the production of L-lysine of a *C. glutamicum* strain comprising a heterologous gene encoding an enzyme with the function of a nicotinamide nucleotide transhydrogenase (PntAB), since PntAB on its own only leads to an increase of the L-lysine yield but has no positive effect on the productivity of said strain.

[0012] Therefore, the present invention concerns a *C. glutamicum* strain having the ability to produce L-lysine, comprising a heterologous gene encoding a NAD(P)(+) transhydrogenase subunit alpha (PntA), a heterologous gene encoding a NAD(P)(+) transhydrogenase subunit beta (PntB) and a gene encoding a myo-inositol permease (IolT) that is overexpressed by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the promoter or by a combination of two or more of these measures.

DETAILED DESCRIPTION OF THE INVENTION

[0013] A heterologous gene means that the gene has been inserted into a host organism which does not naturally have this gene. Insertion of the heterologous gene in the host is performed by recombinant DNA technology. Microorganisms that have undergone recombinant DNA technology are called transgenic, genetically modified or recombinant. Thus, the microorganism, i.e. The *C. glutamicum* strain, according to the present invention is recombinant.

[0014] In the context of the present invention, overexpression may be generally achieved by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the promoter or by a combination of two or more of these measures.

[0015] More particularly, overexpression of a gene can be achieved by increasing the copy number of the gene and/or by an enhancement of regulatory factors, e.g. by functionally linking the gene with a strong promoter and/or by enhancing the ribosomal binding site and/or by codon usage optimization of the start codon or of up to the whole gene. The enhancement of such regulatory factors which positively influence gene expression can, for example, be achieved by modifying the promoter sequence upstream of the structural gene in order to increase the effectiveness of the promoter or by completely replacing said promoter with a more effective or a so-called strong promoter. Promoters are located upstream of the gene. A promoter is a DNA sequence consisting of about 40 to 50 base pairs and which constitutes the binding site for an RNA polymerase holoenzyme and the transcriptional start point, whereby the strength of expression of the controlled polynucleotide or gene can be influenced. Generally, it is possible to achieve an overexpression or an increase in the expression of genes in bacteria by selecting strong promoters, for example by replacing the original promoter with strong, native (originally assigned to other genes) promoters or by modifying certain regions of a given, native promoter (for example its so-called -10 and -35 regions) towards a consensus sequence, e.g. as taught by M. Patek et al. (Microbial Biotechnology 6 (2013), 103-117) for *C. glutamicum*. An example for a "strong" promoter is the superoxide dismutase (sod) promoter ("Psod"; Z. Wang et al., Eng. Life Sci. 2015, 15, 73-82). A "functional linkage" is understood to mean the sequential arrangement of a promoter with a gene, which leads to a transcription of the gene. Another method to increase the expression of a target gene is the optimization of the codon utilization or translocation of the gene onto a more active region of the hosts genome. A further possibility for the overexpression of a gene is the deletion of a repressor or the mutation of the repressor binding region of the promoter, leading to constitutive expression of the gene.

[0016] The *C. glutamicum* strain according to the present invention is able to produce L-lysine. Preferably, the *C. glutamicum* strain according to the present invention is able to excrete L-lysine. L-lysine excreting strains of the species *C. glutamicum* are widely known in the art and can be used for the purpose of the present invention. For example, Blombach et al. (Applied and Environmental Microbiology 75 (2), 419-427, 2009) describe the strain DM1933. Strain DM1933 was obtained from ATCC13032, the *C. glutami-*

cum type strain, by several steps of strain development. Other L-lysine excreting *C. glutamicum* strains are e. g. described in WO 2008033001 A1 and EP 0841395 A1.

[0017] The strain DM2042 is derived from DM1933 (Blombach et al., Applied and Environmental Microbiology 75 (2), 419-427, 2009) with an increased activity of known beneficial lysine production genes *asd*, *dapA*, *dapB*, *lysA*, *lysC* (feedback resistant), *ddh* and *pyc*. These genes are overexpressed by use of a stronger promoter and/or by introduction of additional copies or variants of the respective genes to allow for unlimited flux through the lysine biosynthesis pathway.

[0018] A pure culture of the DM2042 strain has been deposited with the Deutsche Sammlung für Mikroorganismen und Zellkulturen (DSMZ, Braunschweig, Germany) as DSM 34720 in accordance with the Budapest Treaty on 2 Aug. 2023.

[0019] Therefore, in the *C. glutamicum* strain according to the present invention the ability to produce L-lysine is preferably achieved by overexpression of the genes coding for enzymes having the function of aspartate-semialdehyde dehydrogenase (*asd*), aspartate aminotransferase (*aspB*), dihydrodipicolinate synthase (*dapA*), dihydrodipicolinate reductase (*dapB*), diaminopimelate decarboxylase (*lysA*), aspartokinase (*lysC*) and diaminopimelate dehydrogenase (*ddh*), wherein the overexpression is achieved by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the promoter or by a combination of two or more of these measures.

[0020] L-lysine excreting *C. glutamicum* strains typically contain a polynucleotide coding for a feedback resistant aspartokinase polypeptide variant. A feedback resistant aspartokinase polypeptide variant means an aspartokinase which is less sensitive, or desensitized resp., to inhibition by mixtures of L-lysine and L-threonine, e.g. 10 mM each, or mixtures of the L-lysine analogue S-(2-aminoethyl)-L-cysteine and L-threonine, e.g. 50 mM S-(2-aminoethyl)-L-cysteine and 10 mM L-threonine, when compared to the wild form of the enzyme, which is contained in wild type strains such as ATCC13032, ATCC14067 and ATCC13869. The EC number for aspartokinase is EC 2.7.2.4. Descriptions of polynucleotides of *C. glutamicum* encoding a feedback resistant aspartokinase polypeptide variant are given for example in U.S. Pat. Nos. 5,688,671, 6,844,176 and 6,893,848. A summarizing list can be found inter alia in US20090311758A1.

[0021] The *C. glutamicum* strain according to the present invention may further comprise at least one copy of a gene *lysC* coding for a feedback resistant aspartokinase polypeptide variant. Preferably, the feedback resistant aspartokinase polypeptide variant comprises the amino acid sequence according to SEQ ID NO: 27. The amino acid sequence according to SEQ ID NO: 27 differs from the amino acid sequence of the wild type aspartokinase polypeptide of *Corynebacterium glutamicum* ATCC 13032 (the commercially available *C. glutamicum* type strain) in that the amino acid threonine (Thr) at position 311 is replaced by the amino acid isoleucine (Ile).

[0022] The wildtype DNA sequence of the *lysC* gene derived from *Corynebacterium glutamicum* ATCC 13032 encoding for aspartokinase can be achieved under NCBI

Reference Sequence: NC_003450.3, locus_tag CGL_RS01330, also named NCgl0247. The according LysC polypeptide sequence derived from *Corynebacterium glutamicum* ATCC13032 is available under NCBI Reference Sequence: WP_003855724.1.

[0023] The *C. glutamicum* strain of the present invention may further comprise a gene coding for a pyruvate carboxylase (pyc) that is overexpressed by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the promoter or by a combination of two or more of these measures.

[0024] The nucleotide sequence of the *C. glutamicum* ATCC13032 chromosome and its analysis were described by Ikeda and Nakagawa (Applied Microbiology and Biotechnology 62, 99-109 (2003)) and in EP1108790 A2. The information is available at the NCBI under accession number NC_003450.

[0025] The bacteria of the *C. glutamicum* strain of the present invention may comprise in their chromosome a heterologous polynucleotide encoding an amino acid sequence of a NAD(P)(+) transhydrogenase subunit alpha PntA comprising an amino acid sequence according to SEQ ID NO: 3 derived from *Escherichia coli* (*E. coli*) or SEQ ID NO: 7 derived from *Corynebacterium urealyticum* (*C. urealyticum*) and a heterologous polynucleotide encoding an amino acid sequence of a NAD(P)(+) transhydrogenase subunit beta PntB comprising an amino acid sequences according to SEQ ID NO: 4 derived from *E. coli* or SEQ ID NO: 8 derived from *C. urealyticum*.

[0026] The amino acid sequence comparison of PntA from *E. coli* SEQ ID NO: 3 and PntA from *C. urealyticum* SEQ ID NO: 7 results in an identity value of 57%.

[0027] The amino acid sequence comparison of PntB from *E. coli* SEQ ID NO: 4 and PntB from *C. urealyticum* SEQ ID NO: 8 results in an identity value of 66%.

[0028] In the *C. glutamicum* strain according to the present invention the heterologous gene encoding a NAD(P)(+) transhydrogenase subunit alpha is preferably coding for an amino acid sequence that has an identity of at least 60%, preferably of at least 90% to the amino acid sequences according to SEQ ID NO: 7. In a particular embodiment of the present invention the amino acid sequence of the NAD(P)(+) transhydrogenase subunit alpha is identical to the amino acid sequence according to SEQ ID NO: 7.

[0029] In the *C. glutamicum* strain according to the present invention the heterologous gene encoding a NAD(P)(+) transhydrogenase subunit beta is preferably coding for an amino acid sequence that has an identity of at least 70%, preferably of at least 90% to the amino acid sequences according to SEQ ID NO: 8. In a particular embodiment of the present invention the amino acid sequence of the NAD(P)(+) transhydrogenase subunit beta is identical to the amino acid sequence according to SEQ ID NO: 8.

[0030] IolT is a native myo-inositol permease of *C. glutamicum* that is also able to transport glucose into the cell as a side activity but unlike a PTS system, glucose is not phosphorylated by PEP during the transport. *C. glutamicum* harbors two IolT variants, IolT1 and IolT2, which have both been shown to be beneficial. Since both variants are reported to facilitate the same functionality with an amino acid sequence identity of IolT2 to IolT1 of 57%, IolT2 was

chosen to illustrate this invention as a representative for both variants (Lindner, Steffen N., et al., Applied and environmental microbiology, 2011, 77, Nr. 11, S. 3571-3581; Ikeda, Masato, et al. Applied microbiology and biotechnology, 2011, 90, S. 1443-1451; XU, Jian-Zhong, et al., Microbial Cell Factories, 2020, 19, S. 1-15.).

[0031] SEQ ID NO: 11 shows the nucleotide sequence corresponding to iolT2 gene derived from *C. glutamicum* encoding a permease of the major facilitator superfamily (2), NCBI Reference Sequence: NC_003450.3, locus_tag CGL_RS15210, also named NCgl2953.

[0032] SEQ ID NO: 12 shows the IolT2 polypeptide sequence derived from *C. glutamicum*, NCBI Reference Sequence: WP_011015598.1. In the *C. glutamicum* strain according to the present invention the overexpressed gene encoding a myo-inositol permease may encode an amino acid sequence which has an identity of 57% to the amino acid sequence according to SEQ ID NO: 12.

[0033] SEQ ID NO: 28 shows the nucleotide sequence of the iolT1 gene derived from *C. glutamicum* encoding for the permease of the major facilitator superfamily (1), NCBI Reference Sequence: NC_003450.3, locus_tag CGL_RS00945, also named NCgl0178.

[0034] SEQ ID NO: 29 shows the IolT1 polypeptide sequence derived from *C. glutamicum*, GenBank: WP_011013450.1.

[0035] Sequence identity of a conserved polynucleotide or polypeptide is determined by standard alignment algorithms, and may be used with a default gap penalty established by the program used. Substantially homologous or identical sequences are generally capable of hybridizing to all or part of a sequence under moderate or high stringent conditions. It is obvious that hybridization also includes hybridization with polynucleotides containing common codons in polynucleotides or codons taking codon degeneracy into account.

[0036] Whether any two polynucleotide or polypeptide sequences have homology, similarity or identity can be determined, for example, by Pearson et al (1988) [Proc. Natl. Acad. Sci. USA 85]: It can be determined using a known computer algorithm such as the "FASTA" program using default parameters as in 2444. Or, as performed in the Needleman program in the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, Trends Genet. 16:276-277) (version 5.0.0 or later), It can be determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, J. Mol. Biol. 48:443-453) (GCG program package (Devereux, J., et al, Nucleic Acids Research 12:387 (1984)), BLASTP, BLASTN, FASTA (Atschul, [S.] [F.] [ET AL, J MOLEC BIOL 215]: 403 (1990); Guide to Huge Computers, Martin J. Bishop, [ed.,] Academic Press, San Diego, 1994, and [CARILLO ETA/.] (1988) SIAM J Applied Math 48:1073. For example, BLAST from the National Center for Biotechnology Information Database, or Homology, similarity, or identity can be determined using ClustalW.

[0037] Identity of polynucleotides or polypeptides is defined in, for example, Smith and Waterman, Adv. Appl. Math (1981) 2:482, see, e.g., Needleman et al. (1970), J Mol Biol. This can be determined by comparing sequence information using a GAP computer program such as 48:443. In summary, a GAP program can be defined as the total number of symbols in the shorter of the two sequences divided by the number of similarly aligned symbols (i.e., nucleotides or

amino acids). The default parameters for the GAP program are (1) a binary comparison matrix (containing values 1 for identity and 0 for non-identity) and Schwartz and Dayhoff, eds., Atlas Of Protein Sequence And Structure, National Biomedical Research Foundation, pp. 353-358 (1979), Gribkov et al (1986) Nucl. Acids Res. 14: Weighted comparison matrix of 6745 (or EDNAFULL (EMBOSS version of NCBI NUC4.4) permutation matrix); (2) a penalty of 3.0 for each gap and an additional 0.10 penalty for each symbol in each gap (or a gap opening penalty of 10 and a gap extension penalty of 0.5); and (3) no penalty for end gaps. Accordingly, the term "identity" used in this application refers to the relationship between sequences.

[0038] It was found that the *C. glutamicum* strain according to the present invention was able to excrete L-lysine into a suitable medium under suitable fermentation conditions in an increased manner with respect to e.g. product yield (in g L-lysine/l medium or g L-lysine/g carbon source) as well with increased productivity (in concentration of product produced over time in grams product per liter per hour, g/L/h) as compared to the unmodified *C. glutamicum* strain.

[0039] Therefore, the present invention also concerns a method for the fermentative production of L-lysine comprising the steps of cultivating the *C. glutamicum* strain of the present invention and accumulating L-lysine in the medium to form an L-lysine containing fermentation broth.

[0040] The term L-lysine, where mentioned herein, in particular in the context of product formation, also comprises their ionic forms and salts, for example L-lysine mono hydrochloride or L-lysine sulfate. The method according to the present invention may further comprise manufacturing an L-lysine containing product from said fermentation broth or isolating L-lysine from the L-lysine containing fermentation broth. In a fermentative process according to the invention the *C. glutamicum* strain modified in accordance with the present invention, and having the ability to excrete L-lysine is cultivated in a suitable medium under suitable conditions. Due to said ability to excrete said L-lysine the concentration of the L-lysine increases and accumulates in the medium during the fermentative process and the L-lysine is thus produced. Further aspects on fermentation process parameters according to the invention are described e.g. in EP3686277 A1.

SHORT DESCRIPTION OF THE SEQUENCES

[0041] SEQ ID NO: 1 DNA sequence of the pntA gene derived from *Escherichia coli* encoding the pyridine nucleotide transhydrogenase subunit alpha, pntA: NCBI-GeneID: 946628.

[0042] SEQ ID NO: 2 DNA sequence of the pntB gene derived from *Escherichia coli* encoding the pyridine nucleotide transhydrogenase subunit beta, pntB: NCBI-GeneID: 946144.

[0043] SEQ ID NO: 3 PntA polypeptide sequence derived from *Escherichia coli*, NCBI Reference Sequence: PntA: NP_416120.1.

[0044] SEQ ID NO: 4 PntB polypeptide sequence derived from *Escherichia coli*, NCBI Reference Sequence: PntB: NP_416119.1.

[0045] SEQ ID NO: 5 DNA sequence of the pntA gene derived from *Corynebacterium urealyticum* encoding the NAD(P)(+) transhydrogenase subunit alpha, NCBI Reference Sequence: pntA: CKV82_RS04765.

[0046] SEQ ID NO: 6 DNA sequence of the pntB gene derived from *Corynebacterium urealyticum* encoding the NAD(P)(+) transhydrogenase subunit beta, NCBI Reference Sequence: pntB: CKV82_RS04770.

[0047] SEQ ID NO: 7 PntA polypeptide sequence derived from *Corynebacterium urealyticum*, NCBI Reference Sequence: PntA: WP_012360758.1.

[0048] SEQ ID NO: 8 PntB polypeptide sequence derived from *Corynebacterium urealyticum*, NCBI Reference Sequence: PntB: WP_012360759.1.

[0049] SEQ ID NO: 9 pntAB_ec polynucleotide carrying intergenic sequences derived from *Corynebacterium glutamicum*, promotor P2 (base pair position 807-986), codon optimized variant of pntA and pntB genes derived from *Escherichia coli*.

[0050] SEQ ID NO: 10 pntAB_cu polynucleotide carrying intergenic sequences derived from *Corynebacterium glutamicum*, promotor P2 (base pair position 807-986), pntA and pntB genes derived from *Corynebacterium urealyticum*.

[0051] SEQ ID NO: 11 DNA sequence of the iolT2 gene derived from *Corynebacterium glutamicum* encoding for a permease of the major facilitator superfamily 2, NCBI Reference Sequence: NC_003450.3, locus_tag CGL_RS15210, also named NCgl2953

[0052] SEQ ID NO: 12 IolT2 polypeptide sequence derived from *Corynebacterium glutamicum*, NCBI Reference Sequence: WP_011015598.1.

[0053] SEQ ID NO: 13 P1-iolT2 polynucleotide carrying intergenic sequences, promotor P1 (base pair position 792-911) and initial sequence of the iolT2 gene derived from *Corynebacterium glutamicum*.

[0054] SEQ ID NO: 14 DNA sequence of the ptsG gene derived from *Corynebacterium glutamicum* encoding for the glucose-specific phosphotransferase system enzyme II, GenBank: CGL_RS06785.

[0055] SEQ ID NO: 15 PtsG polypeptide sequence derived from *Corynebacterium glutamicum*, GenBank: ABB53258.1.

[0056] SEQ ID NO: 16 P2-ptsG polynucleotide carrying intergenic sequences, promotor P2 (base pair position 751-930) and initial sequence of the ptsG gene derived from *Corynebacterium glutamicum*.

[0057] SEQ ID NO: 17 DNA sequence of the glf gene derived from *Zymomonas mobilis* encoding a glucose transport protein, GenBank: M60615.1, base pair position 185-1606

[0058] SEQ ID NO: 18 Glf polypeptide sequence derived from *Zymomonas mobilis*, Proteine ID: AAA27691.1

[0059] SEQ ID NO: 19 P1-glf polynucleotide carrying intergenic sequences derived from *Corynebacterium glutamicum*, promotor P1 (base pair position 792-911) and glf gene derived from *Zymomonas mobilis*.

[0060] SEQ ID NO: 20 DNA sequence of the galP gene derived from *Escherichia coli* encoding a galactose: H (+) symporter, Gene ID: 947434.

[0061] SEQ ID NO: 21 GalP polypeptide sequence derived from *Escherichia coli*, NCBI Reference Sequence: NP_417418.1

[0062] SEQ ID NO: 22 P1-galP polynucleotide carrying intergenic sequences derived from *Corynebacterium glutamicum*, promotor P1 (base pair position 792-911) and galP gene derived from *Escherichia coli*.

[0063] SEQ ID NO: 23 DNA sequence of the bgf gene derived from *Corynebacterium glutamicum* R encoding Ell

component of the β -glucoside-PTS, GenBank: AF508972.1, base pair position 1320-3176.

[0064] SEQ ID NO: 24 BglF polypeptide sequence derived from *Corynebacterium glutamicum* R, GenBank: AAO15901.1.

[0065] SEQ ID NO: 25 P1-bglF polynucleotide carrying intergenic sequences derived from *Corynebacterium glutamicum*, promotor P1 (base pair position 792-911) and bglF gene derived from *Corynebacterium glutamicum* R.

[0066] SEQ ID NO: 26 DNA sequence of the lysC gene derived from *Corynebacterium glutamicum* encoding the feedback resistant aspartokinase polypeptide variant LysC_T311I

[0067] SEQ ID NO: 27 LysC_T311I polypeptide sequence derived from *Corynebacterium glutamicum*,

[0068] SEQ ID NO: 28 DNA sequence of the ioIT1 gene derived from *Corynebacterium glutamicum* encoding for the permease of the major facilitator superfamily 1, NCBI Reference Sequence: NC_003450.3, locus_tag CGL_RS00945, also named NCgl0178

[0069] SEQ ID NO: 29 IoIT1 polypeptide sequence derived from *Corynebacterium glutamicum*, GenBank: WP_011013450.1

[0070] The term of intergenic sequences used herein implies homologous sequences which are used to enable the replacement of native promotor regions or to incorporate heterologous genes by homologous recombination.

EXPERIMENTAL SECTION

A) Materials and Methods

[0071] The molecular biology kits, primers and chemicals used and some details of the methods applied are briefly described herewith.

1. Antibiotics and Chemicals

[0072] a. Kanamycin: Kanamycin solution from *Streptomyces kanamyceticus* from Sigma Aldrich (St. Louis, USA, Cat. no. K0254).

[0073] b. Nalidixic acid: Nalidixic acid sodium salt from Sigma Aldrich (St. Louis, USA, Cat. no. N4382).

[0074] c. If not stated otherwise, all chemicals were purchased analytically pure from Merck (Darmstadt, Germany), Sigma Aldrich (St. Louis, USA) or Carl-Roth (Karlsruhe, Germany).

2. Cultivation

[0075] If not stated otherwise, all cultivation/incubation procedures were performed as follows herewith:

[0076] a. LB broth (MILLER) from Merck (Darmstadt, Germany; Cat. no. 110285) was used to cultivate *E. coli* strains in liquid medium. The liquid cultures (10 ml liquid medium per 100 ml Erlenmeyer flask with 3 baffles) were incubated in the Infors HT Multitron standard incubator shaker from Infors GmbH (Einsbach, Germany) at 37° C. and 200 rpm.

[0077] b. LB agar (MILLER) from Merck (Darmstadt, Germany Cat. no. 110283) was used for cultivation of *E. coli* strains on agar plates. The agar plates were incubated at 37° C. in an INCU-Line® mini-incubator from VWR (Radnor, USA).

[0078] c. Brain heart infusion broth (BHI) from Merck (Darmstadt, Germany; Cat. no. 110493) was used to culti-

vate *C. glutamicum* strains in liquid medium. The liquid cultures (10 ml liquid medium per 100 ml Erlenmeyer flask with 3 baffles) were incubated in the Infors HT Multitron standard incubator shaker from Infors GmbH (Einsbach, Germany) at 33° C. and 200 rpm.

[0079] d. Brain heart agar (BHI-agar) from Merck (Darmstadt, Germany; Cat. no. 113825) was used for cultivation of *C. glutamicum* strains on agar plates. The agar plates were incubated at 33° C. in an incubator from Heraeus Instruments with Kelvitron® temperature controller (Hanau, Germany).

3. Determining Optical Density

[0080] a. The optical density of bacterial suspensions in shake flask cultures was determined at 600 nm (OD600) using the BioPhotometer from Eppendorf AG (Hamburg, Germany).

[0081] b. The optical density of bacterial suspensions produced in the Wouter Duetz (WDS) micro fermentation system (24-Well Plates) was determined at 660 nm (OD660) with the GENios™ plate reader from Tecan Group AG (Mannedorf, Switzerland).

4. Centrifugation

a. Benchtop Centrifuge for Reaction Tubes with A Volume Up to 2 ml

[0082] Bacterial suspensions with a maximum volume of 2 ml were caused to sediment using 1 ml or 2 ml reaction tubes (e.g., Eppendorf Tubes® 3810X) using an Eppendorf 5417 R centrifuge (5 min at 13.000 rpm).

b. Benchtop Centrifuge for Tubes with a Volume Up to 50 ml

[0083] Bacterial suspensions with a maximum volume of 50 ml were caused to sediment using 15 ml or 50 ml centrifuge tubes (e.g. Falcon™ 50 ml Conical Centrifuge Tubes) using an Eppendorf 5810 R centrifuge for 10 min at 4.000 rpm.

5. PCR

[0084] Integration of desired DNA fragments into the chromosome of *C. glutamicum* was detected by PCR amplification and following Sanger sequencing of the PCR product.

a. Primers

[0085] The oligonucleotides used were synthesized by eurofins genomics GmbH (Ebersberg, Germany).

b. Template

[0086] As PCR template the total DNA contained in a colony was used. It was prepared by taking cell material with a toothpick from a colony on an agar plate and placing the cell material directly into the PCR reaction tube. The cell material was heated for 10 sec with 800 W in a microwave oven type Mikrowave & Grill from SEVERIN Elektrogeräthe GmbH (Sundern, Germany) and then the PCR reagents were added to the template in the PCR reaction tube.

b. Reaction Mix

[0087] The PCR reaction was carried out using SapphireAmp® Fast PCR Master Mix from Takara Bio Inc. (Shiga, Japan). The reaction mixture was prepared according to the manufacturer's information. Therefore 25 µl of the SapphireAmp® Fast PCR Master Mix (2×) was mixed with 0.5 µl of each of the PCR-Primers [100 pmol/µl] and 24 µl

H₂O to get in total 50 µl reaction volume. The thermocycling conditions for PCR are shown in Table 1.

TABLE 1

Thermocycling conditions for PCR PCR-program			
Step	Time [sec]	T [° C.]	Description
1	60	94	Denaturation
2	05	98	Amplification
3	05	55	Denaturation step
4	10 sec/kb	72	Annealing step
			Elongation step
			Repeat step 2 to 4: 30 x

c. PCR Cycler

[0088] The reactions were carried out in a Mastercycler® of Eppendorf (Hamburg, Germany).

6. Chemical Transformation of *E. coli*

[0089] *E. coli* K-12 strain S17-1 was used as donor for conjugational transfer of plasmids based on pK18mobsacB from *E. coli* to *C. glutamicum*. Strain S17-1 is described by Simon, R. et al. (BiolTechnology 1, 784-794, 1983). It is available from the American Type Culture Collection under the access number ATCC47055.

[0090] Chemically competent *E. coli* S17-1 cells were made as follows: A preculture of 10 ml LB medium (10 ml liquid medium per 100 ml Erlenmeyer flask with 3 baffles) was inoculated with 100 µl bacterial suspension of strain S17-1 and the culture was incubated overnight for about 18 h at 37° C. and 250 rpm. The main culture (70 ml LB contained in a 250 ml Erlenmeyer flask with 3 baffles) was inoculated with 300 µl of the preculture and incubated up to an OD₆₀₀ of 0.5-0.8 at 37° C. The culture was centrifuged for 6 min at 4° C. and 4000 rpm and the supernatant was discarded. The cell pellet was resuspended in 20 ml sterile, ice-cold 50 mM CaCl₂ solution and incubated on ice for 30 min. After another centrifugation step, the pellet was resuspended in 5 ml ice-cold 50 mM CaCl₂ solution and the suspension incubated on ice for 30 min. The cell suspension was then adjusted to a final concentration of 20% glycerol (v/v) with 85% (v/v) sterile ice-cold glycerol. The suspension was divided into 50 µl aliquots and stored at -80° C.

[0091] To transform S17-1 cells, the protocol according to Tang et al. (Nucleic Acids Res. 22(14), 2857-2858, 1994) with a heat shock of 45 sec. was used.

7. Conjugation of *C. glutamicum*

[0092] The pK18mobsacB plasmid system described by Schäfer et al. (Gene 145, 69-73, 1994) was used to integrate desired DNA fragments into the chromosome of *C. glutamicum*. A modified conjugation method of Schäfer et al. (Journal of Bacteriology 172, 1663-1666, 1990) was used to transfer the respective plasmid into the desired *C. glutamicum* recipient strain.

[0093] Liquid cultures of the *C. glutamicum* strains were carried out in BHI medium at 33° C. The heat shock was carried out at 48.5° C. for 9 min. Transconjugants were selected by plating the conjugation batch on EM8 agar (Table 2), which was supplemented with 25 mg/l kanamycin and 50 mg/l nalidixic acid. The EM8 agar plates were incubated for 72 h at 33° C.

TABLE 2

Composition of the EM8 agar	
Components	Concentration (g/l)
Glucose (sterile filtered)	23
CSL (corn steep liquor; Roquette; solid content 48 ± 2% w/w)	30
Peptone from soymeal (Merck, Germany)	40
(NH ₄) ₂ SO ₄	8
Urea	3
KH ₂ PO ₄	4
MgSO ₄ •7 H ₂ O	0.5
FeSO ₄ •7 H ₂ O	0.01
CuSO ₄ •5 H ₂ O	0.001
ZnSO ₄ •7 H ₂ O	0.01
Calcium pantothenate, D(+)	0.01
Thiamine	0.001
Inositol	0.1
Nicotinic acid	0.001
Biotin (sterile filtered)	0.005
CaCO ₃ (autoclaved separately)	1.6
Agar-Agar (Merck, Germany)	14

[0094] Sterile toothpicks were used to transfer the transconjugants onto BHI agar, which was supplemented with 25 mg/l kanamycin and 50 mg/l nalidixic acid. The agar plates were incubated for 20 h at 33° C. The cultures of the respective transconjugants produced in this manner were then propagated further for 24 h at 33° C. in 10 ml BHI medium contained in 100 ml Erlenmeyer flasks with 3 baffles. An aliquot was taken from the liquid culture suitably diluted and plated (typically 100 to 200 µl) on BHI agar which was supplemented with 10% saccharose. The agar plates were incubated for 48 h at 33° C. The colonies growing on the saccharose containing agar plates were then examined for the phenotype kanamycin sensitivity. To do so a toothpick was used to remove cell material from the colony and to transfer it onto BHI agar containing 25 mg/l kanamycin and onto BHI agar containing 10% saccharose. The agar plates were incubated for 60 h at 33° C. Clones that proved to be sensitive to kanamycin and resistant to saccharose were examined for integration of the desired DNA fragment by means of PCR and following sequencing of the PCR product.

8. Glycerol Stocks of *E. coli* and *C. glutamicum* Strains

[0095] For long time storage of *E. coli*- and *C. glutamicum* strains glycerol stocks were prepared. Selected *E. coli* clones were cultivated in 10 ml LB medium supplemented with 2 g/l glucose. Selected *C. glutamicum* clones were cultivated in two-fold concentrated BHI medium supplemented with 2 g/l glucose. Cultures of plasmid containing *E. coli* strains were supplemented with 50 mg/l kanamycin. Cultures of plasmid containing *C. glutamicum* strains were supplemented with 25 mg/l kanamycin. The medium was contained in 100 ml Erlenmeyer flasks with 3 baffles. It was inoculated with a loop of cells taken from a colony and the culture incubated for about 18 h at 37° C. and 200 rpm in the case of *E. coli* and 33° C. and 200 rpm in the case of *C. glutamicum*. After said incubation period 1.2 ml 85% (v/v) sterile glycerol were added to the culture. The obtained glycerol containing cell suspension was then aliquoted in 2 ml portions and stored at -80° C.

9. Cultivation System According to Wouter Duetz (WDS)

[0096] The milliliter-scale cultivation system according to Duetz (Trends Microbiol. 2007; 15 (10): 469-75) was used

to investigate the performance of the *C. glutamicum* strains constructed. For this purpose, 24-deepwell microplates (24 well WDS plates) from EnzyScreen BV (Heemstede, Netherlands; Cat. no. CR1424), filled with 2.5 ml medium were used.

[0097] Precultures of the strains were done in 10 ml two-fold concentrated BHI medium. The medium was contained in a 100 ml Erlenmeyer flask with 3 baffles. It was inoculated with 100 µl of a glycerol stock culture and the culture incubated for 24 h at 33° C. and 200 rpm.

[0098] After said incubation period the optical densities OD600 of the precultures were determined.

[0099] The main cultures were done by inoculating the 2.5 ml medium containing wells of the 24 Well WDS-Plate with an aliquot of the preculture to give an optical density OD600 of 0.1.

[0100] As medium for the main culture CGXII medium (described by Keilhauer et al. (J. Bacteriol. 1993 September; 175 (17): 5595-5603)) with 7.5 g/l CSL and 20 g/l glucose was used. The composition of the modified CGXII medium is shown in Table 3.

TABLE 3

Composition of Keilhauer's CGXII medium (modified)	
Components	Concentration (g/l)
MOPS (3-(N-Morpholino)propanesulfonic acid)	42
(NH ₄) ₂ SO ₄	20
Urea	5
KH ₂ PO ₄	1
K ₂ HPO ₄	1
MgSO ₄ •7 H ₂ O	0.25
CaCl ₂	0.01
FeSO ₄ •7 H ₂ O	0.01
MnSO ₄ H ₂ O	0.01
ZnSO ₄ •7 H ₂ O	0.001
CuSO ₄ •5 H ₂ O	0.0002
NiCl ₂ 6 H ₂ O	0.00002
Biotin (sterile-filtered)	0.0002
Protocatechuic acid (sterile-filtered)	0.03
CSL (corn steep liquor; Roquette; solid content 50% w/w)	7.5
Glucose (sterile filtered)	20.0
adjust the pH to 7 with NaOH	

[0101] These main cultures were incubated for approximately 45 h at 33° C. and 300 rpm in an Infors HT Multitron standard incubator shaker from Infors GmbH (Bottmingen, Switzerland) until complete consumption of glucose.

[0102] The glucose concentration in the suspension was analyzed with the blood glucose-meter OneTouch Vita® from LifeScan (Johnson & Johnson Medical GmbH, Neuss, Germany).

[0103] After cultivation the culture suspensions were transferred to a deep well microplate. A part of the culture suspension was suitably diluted to measure the OD660. Another part of the culture was centrifuged and the concentration of L-amino acids, in particular L-lysine, and residual glucose were analyzed in the supernatant.

10. Amino Acid Analyser

[0104] The concentration of L-lysine and other L-amino acids, e.g. L-valine, in the culture supernatants was determined by ion exchange chromatography using a SYKAM S433 amino acid analyser from SYKAM Vertriebs GmbH

(Fürstenfeldbruck, Germany). As solid phase a column with spherical, polystyrene-based cation exchanger (Peek LCA N04/Na, dimension 150x4.6 mm) from SYKAM was used. Depending on the L-amino acid the separation takes place in an isocratic run using a mixture of buffers A and B for elution or by gradient elution using said buffers. As buffer A an aqueous solution containing in 20 l 263 g trisodium citrate, 120 g citric acid, 1100 ml methanol, 100 ml 37% HCl and 2 ml octanoic acid (final pH 3.5) was used. As buffer B an aqueous solution containing in 20 l 392 g trisodium citrate, 100 g boric acid and 2 ml octanoic acid (final pH 10.2) was used. The free amino acids were coloured with ninhydrin through post-column derivatization and detected photometrically at 570 nm.

11. Glucose Determination with Continuous Flow System (CFS)

[0105] A SANplus multi-channel continuous flow analyser from SKALAR analytic GmbH (Erkelenz, Germany) was used to determine the concentration of glucose in the supernatant. Glucose was detected with a coupled-enzyme assay (Hexokinase/Glucose-6-Phosphate-Dehydrogenase) via NADH formation.

B) Experimental Results

Example 1

Sequence of the pntA and pntB Genes Derived from *Escherichia coli* and pntA and pntB Genes Derived from *Corynebacterium urealyticum*

[0106] The pntA (SEQ ID NO: 1) and pntB (SEQ ID NO: 2) genes derived from *Escherichia coli* code for a pyridine nucleotide transhydrogenase subunit alpha and pyridine nucleotide transhydrogenase subunit beta, respectively. The encoded polypeptide sequence can be found at the NCBI under NCBI Reference Sequence: pntA: NP_416120.1 (SEQ ID NO: 3), pntB: NP_416119.1 (SEQ ID NO: 4).

[0107] The pntA (NCBI Reference Sequence: CKV82_RS04765) (SEQ ID NO: 5) and pntB (NCBI Reference Sequence: CKV82_RS04770) (SEQ ID NO: 6) genes derived from *Corynebacterium urealyticum* code for a NAD (P)(+) transhydrogenase subunit alpha and NAD(P)(+) transhydrogenase subunit beta, respectively. The encoded polypeptide sequence can be found at the NCBI under the description Re/Si-specific NAD(P)(+) transhydrogenase subunit alpha as NCBI Reference Sequence: pntA: WP_012360758.1 (SEQ ID NO: 7) and pntB: WP_012360759.1 (SEQ ID NO: 8) under the description NAD(P)(+) transhydrogenase (Re/Si-specific) subunit beta.

Example 2

Construction of Plasmids pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu

[0108] Plasmids pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu were constructed to enable incorporation of the heterologous pntAB genes pntAB_ec and pntAB_cu in lysine producing strain DM2042. The plasmids are based on the mobilizable vector pK18mobsacB described by Schäfer et al. (Gene 145, 69-73, 1994). For construction of pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu, the pntAB_ec polynucleotide according to SEQ ID NO: 9 (carrying intergenic sequences, promotor P2, pntA gene and pntB gene derived from *Escherichia coli*) and the pntAB_cu polynucleotide accord-

ing to SEQ ID NO: 10 (carrying intergenic sequences, promotor P2, pntA gene and pntB gene derived from *Corynebacterium urealyticum*) were synthesized and sub-cloned into pK18mobsacB by GeneArt (ThermoFisher Scientific (Waltham, USA)).

[0109] The intergenic sequences of SEQ ID NO: 9 and SEQ ID NO: 10 derived from *Corynebacterium glutamicum* represent homologous sequences which enable to incorporate the heterologous pntA and pntB genes by homologous recombination.

[0110] Seq ID NO: 9 represents the codon optimized variant of pntA and pntB gene sequences which were used for construction the plasmid pK18mobsacB_pntAB_ec.

[0111] Seq ID NO: 10 comprises an ATG nucleotide as start codon which is a variant of the native pntA gene sequence derived from *Corynebacterium urealyticum* which was used for construction the plasmid pK18mobsacB_pntAB_cu.

[0112] To assemble the plasmids pK18mobsacB_pntAB_ec the following steps were done by GeneArt: The two polynucleotides i.e., the vector pK18mobsacB and the polynucleotide pntAB_ec were treated with XmaI and SbfI, ligated and the ligation mixture used to transform *E. coli*.

[0113] To assemble the plasmids pK18mobsacB_pntAB_cu the following steps were done by GeneArt: The two polynucleotides i.e., the vector pK18mobsacB and the polynucleotide pntAB_cu were treated with XbaI and SbfI, ligated and the ligation mixture used to transform *E. coli*.

[0114] DNA of plasmids pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu was isolated from a transformant and the polynucleotides pntAB_ec and pntAB_cu, respectively, created within pK18mobsacB was analyzed by Sanger sequencing.

Example 3

Construction of Strains DM2042_pntAB_ec and DM2042_pntAB_cu

[0115] Strain DM1933 is an L-lysine producer derived from the *C. glutamicum* wild type strain ATCC 13032 and has been described by Blombach et al. (Applied and Environmental Microbiology 75 (2), 419-427, 2009).

[0116] Strain DM2042 is derived from DM1933, with an increased activity of known beneficial lysine production

genes asd, dapA, dapB, lysA, lysC (feedback resistant), ddh and pyc. These genes are overexpressed by use of a stronger promotor and/or introduction of additional copies or variants of the respective gene to allow for unlimited flux through the lysine biosynthesis pathway.

[0117] A pure culture of the DM2042 strain has been deposited with the Deutsche Sammlung für Mikroorganismen und Zellkulturen (DSMZ, Braunschweig, Germany) as DSM 34720 in accordance with the Budapest Treaty on 2 Aug. 2023.

[0118] The plasmids pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu obtained in example 2 were used to incorporate the heterologous genes pntAB_ec derived from *E. coli* and pntAB_cu derived from *C. urealyticum*, respectively.

[0119] Chemically competent cells of *E. coli* strain S17-1 were transformed with plasmid DNA of pK18mobsacB_pntAB_ec and pK18mobsacB_pntAB_cu. The presence of the plasmids obtained in example 2 were analyzed by Sanger sequencing, respectively.

[0120] For conjugal transfer into the strain DM2042 the modified conjugation method of Schäfer et al. (Journal of Bacteriology 172, 1663-1666, 1990) as described in materials and methods was used. Subsequently selection of transconjugant clones was done by identification of saccharose resistance and kanamycin sensitivity phenotype.

[0121] Transconjugant clones of DM2042 carrying the heterologous pntAB genes pntAB_ec or pntAB_cu were selected according to chromosome sequence analysis. The strain carrying the heterologous pntAB_ec gene was called DM2042_pntAB_ec. The strain carrying the heterologous pntAB_cu gene was called DM2042_pntAB_cu.

[0122] A glycerol stock culture of the transconjugant clone was prepared and used as starting material for further investigations.

Example 4

Sequence of iolT2 and ptsG Genes Derived from *Corynebacterium glutamicum*.

[0123] The nucleotide sequence of iolT2 (CGL_RS15210) and ptsG (CGL_RS06785) genes derived from *Corynebacterium glutamicum* as well as the encoded polypeptide sequences can be found at the NCBI. The corresponding NCBI Reference Sequences are listed below in Table 4.

TABLE 4

iolT2 and ptsG genes derived from <i>Corynebacterium glutamicum</i>							
gene				polypeptide sequences			
				NCBI reference			
name	SEQ ID	NCBI / GenBank ID		name	SEQ ID	sequence / GenBank ID	function organism
iolT2	11	NC_003450.3, locus_tag CGL_RS15210		IolT2	12	WP_011015598.1	permease of the major facilitator superfamily <i>C. glutamicum</i>
ptsG	14	NC_003450.3, locus_tag CGL_RS06785		PtsG	15	WP_011014304.1	glucose-specific phosphotransferase system enzyme II <i>C. glutamicum</i>

Example 5

Construction of Plasmids pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG*

[0124] Plasmids pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG* were constructed to enable replacement of the nucleotide sequence upstream to *iolT2* and *ptsG* respectively with the sequence of promoter P1 and P2 in lysine producing strains DM2042, DM2042_pntAB_ec and DM2042_pntAB_cu. The plasmids are based on the mobilizable vector pK18mobsacB described by Schäfer et al. (Gene 145, 69-73, 1994). For construction of pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG*, the P1-*iolT* polynucleotide according to SEQ ID NO: 13 (carrying intergenic sequences, promoter P1 and *iolT2* gene derived from *Corynebacterium glutamicum*) and the P2-*ptsG* polynucleotide according to SEQ ID NO: 16 (carrying intergenic sequences, promoter P2 and *ptsG* gene derived from *Corynebacterium glutamicum*) were synthesized and subcloned into pK18mobsacB by GeneArt (ThermoFisher Scientific (Waltham, USA)).

[0125] The intergenic sequences of SEQ ID NO: 13 and SEQ ID NO: 16 derived from *Corynebacterium glutamicum* represent homologous sequences which enable to replace the native promoter region of *iolT2* and *ptsG* respectively by homologous recombination.

[0126] To assemble the plasmids pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG* the following steps were done by GeneArt: The two polynucleotides i.e., the vector pK18mobsacB and the corresponding polynucleotide P1-*iolT* and P2-*ptsG* were treated with 2 restriction enzymes (see Table 5 below), ligated and the ligation mixture used to transform *E. coli*.

[0127] DNA of plasmids pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG* were isolated from transformants and the polynucleotide P1-*iolT* and P2-*ptsG* created within pK18mobsacB were analyzed by Sanger sequencing.

TABLE 5

Construction of plasmids pK18mobsacB_P1- <i>iolT</i> and pK18mobsacB_P2- <i>ptsG</i>			
polynucleotide		construction	
name	SEQ ID	restriction enzymes	construct
P1- <i>iolT</i>	13	BamHI, SbfI	pK18mobsacB_P1- <i>iolT</i>
P2- <i>ptsG</i>	16	XbaI, SbfI	pK18mobsacB_P2- <i>ptsG</i>

Example 6

Construction of Strains DM2042_P1-*iolT*, DM2042_pntAB_ec_P1-*iolT*, DM2042_pntAB_cu_P1-*iolT*, DM2042_P2-*ptsG* and DM2042_pntAB_cu_P2-*ptsG*

[0128] The plasmids pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG* obtained in example 5 were used to replace the native promoter regions upstream of the *iolT2* gene derived from *C. glutamicum* and the *ptsG* gene derived from *C. glutamicum*, respectively.

[0129] Chemically competent cells of *E. coli* strain S17-1 were transformed with plasmid DNA of pK18mobsacB_P1-*iolT* and pK18mobsacB_P2-*ptsG*. The presence of the plasmids obtained in example 5 were analyzed by Sanger sequencing, respectively.

[0130] For conjugal transfer into the strains DM2042, DM2042_pntAB_ec and DM2042_pntAB_cu the modified conjugation method of Schäfer et al. (Journal of Bacteriology 172, 1663-1666, 1990) as described in materials and methods was used. Subsequently, selection of transconjugant clones was done by identification of saccharose resistance and kanamycin sensitivity phenotype. Transconjugant clones of DM2042, DM2042_pntAB_ec and DM2042_pntAB_cu carrying the replaced P1 Promotor upstream to *iolT2* were selected according to chromosome sequence analysis. The respective strains carrying the replaced P1 Promotor upstream to *iolT2* were called DM2042_P1-*iolT*, DM2042_pntAB_ec_P1-*iolT* and DM2042_pntAB_cu_P1-*iolT*.

[0131] Transconjugant clones of DM2042 and DM2042_pntAB_cu carrying the replaced P2 Promotor upstream to *ptsG* were selected according to chromosome sequence analysis. The respective strains carrying the replaced P2 Promotor upstream to *ptsG* were called DM2042_P2-*ptsG* and DM2042_pntAB_cu_P2-*ptsG*.

[0132] Glycerol stock cultures of the transconjugant clones were prepared and used as starting material for further investigations.

Example 7

Sequence of *Glf* Gene Derived from *Zymomonas mobilis*, *galP* Gene Derived from *Escherichia coli* and *bglF* Gene Derived from *Corynebacterium glutamicum* R.

[0133] The nucleotide sequence of the *glf* gene derived from *Zymomonas mobilis*, the *galP* gene derived from *Escherichia coli* and *bglF* gene derived from *Corynebacterium glutamicum* R as well as the encoded polypeptide sequences can be found at the NCBI. The corresponding NCBI Reference Sequences are listed below in Table 6.

TABLE 6

glf gene derived from <i>Zymomonas mobilis</i> , galP gene derived from <i>Escherichia coli</i> and bglF gene derived from <i>Corynebacterium glutamicum</i> R.								
polypeptide sequences								
gene				NCBI Reference Sequence /				
name	SEQ ID	NCBI / GenBank / ID	GenBank /	name	SEQ ID	GenBank / Protein ID /	function	organism
glf	17	M60615.1, base pair position 185-1606		Glf	18	AAA27691.1	glucose transport protein	<i>Zymomonas mobilis</i>

TABLE 6-continued

glf gene derived from <i>Zymomonas mobilis</i> , galP gene derived from <i>Escherichia coli</i> and bglF gene derived from <i>Corynebacterium glutamicum</i> R.								
			polypeptide sequences					
gene			NCBI Reference Sequence /					
name	SEQ ID	NCBI / GenBank / Gen ID	name	SEQ ID	GenBank / Protein ID /	function	organism	
galP	20	947434	GalP	21	NP_417418.1	galactose:H(+) symporter	<i>E. coli</i>	
bglF	23	AF508972.1, base pair position 1320-3176	BglF	24	AAO15901.1	EII component of the β-glucoside-PTS	<i>C. glutamicum</i> R	

Example 8

Construction of Plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR

[0134] Plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR were constructed to enable incorporation of the heterologous glf gene, galP gene and bglF gene in lysine producing strain DM2042_pntAB_cu. The plasmids are based on the mobilizable vector pK18mobsacB described by Schäfer et al. (Gene 145, 69-73, 1994).

[0135] For construction of pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR, the P1-glf_zm polynucleotide according to SEQ ID NO: 19 (carrying intergenic sequences, promotor P1 and glf gene derived from *Zymomonas mobilis*), the P1-galP_ec polynucleotide according to SEQ ID NO: 22 (carrying intergenic sequences, promotor P1 and galP gene derived from *Escherichia coli*) and the P1-bglF_cgR polynucleotide according to SEQ ID NO: 25 (carrying intergenic sequences, promotor P1 and bglF gene derived from *Corynebacterium glutamicum* R) were synthesized and subcloned into pK18mobsacB by GeneArt (ThermoFisher Scientific (Waltham, USA)).

[0136] The intergenic sequences of SEQ ID NO: 17, SEQ ID NO: 20 and SEQ ID NO: 23 derived from *Corynebacterium glutamicum* represent homologous sequences which enable to incorporate the heterologous glf, galP and bglF genes by homologous recombination.

[0137] To assemble the plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR the following steps were done by GeneArt: The two polynucleotides i.e., the vector pK18mobsacB and the corresponding polynucleotide P1-glf_zm, P1-galP_ec and P1-bglF_cgR were treated with 2 restriction enzymes (see Table 7 below), ligated and the ligation mixture used to transform *E. coli*.

[0138] DNA of plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR were isolated from transformants and the polynucleotide P1-glf_zm, P1-galP_ec and P1-bglF_cgR created within pK18mobsacB were analyzed by Sanger sequencing.

TABLE 7

Construction of plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR			
construction			
polynucleotide	restriction		
name	SEQ ID	enzymes	construct
P1-glf_zm	19	XmaI, SbfI	pK18mobsacB_P1-glf_zm
P1-galP_ec	22	XmaI, SbfI	pK18mobsacB_P1-galP_ec
P1-bglF_cgR	25	XmaI, SbfI	pK18mobsacB_P1-bglF_cgR

Example 9

Construction of Strain DM2042_pntAB_cu_P1-glf_zm, DM2042_pntAB_cu_P1-galP_ec and DM2042_pntAB_cu_P1-bglF_cgR

[0139] The plasmids pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR obtained in example 8 were used to incorporate the heterologous gene glf_zm derived from *Zymomonas mobilis*, galP_ec derived from *E. coli* and bglF_cgR derived from *Corynebacterium glutamicum* R, respectively.

[0140] Chemically competent cells of *E. coli* strain S17-1 were transformed with plasmid DNA of pK18mobsacB_P1-glf_zm, pK18mobsacB_P1-galP_ec and pK18mobsacB_P1-bglF_cgR. The presence of the plasmids obtained in example 8 were analyzed by Sanger sequencing, respectively.

[0141] For conjugal transfer into the strain DM2042_pntAB_cu the modified conjugation method of Schäfer et al. (Journal of Bacteriology 172, 1663-1666, 1990) as described in materials and methods was used. Subsequently selection

of transconjugant clones was done by identification of saccharose resistance and kanamycin sensitivity phenotype.

[0142] Transconjugant clones of DM2042_pntAB_cu carrying the heterologous glf_zm gene, galP_ec gene or bglF_cgR gene were selected according to chromosome sequence analysis. The strains carrying the heterologous glf_zm gene was called and DM2042_pntAB_cu_P1-glf_zm. The strain carrying the heterologous galP_ec gene was called DM2042_pntAB_cu_P1-galP_ec. The strain carrying the heterologous bglF_cgR gene was called DM2042_pntAB_cu_P1-bglF_cgR.

[0143] A glycerol stock culture of the transconjugant clones was prepared and used as starting material for further investigations.

Example 10

L-Lysine Production and Productivity

[0144] Strains DM2042 (reference), DM2042_P1-ioIT carrying the genetically modified promotor P1 upstream to ioIT2, DM2042_pntAB_ec carrying the pntAB_ec genes derived from *E. coli*, DM2042_pntAB_ec_P1-ioIT carrying the pntAB_ec genes derived from *E. coli* and the genetically modified promotor P1 upstream to ioIT2, strain DM2042_pntAB_cu carrying the pntAB_cu genes derived from *C. urealyticum* and strain DM2042_pntAB_cu carrying the pntAB_cu genes derived from *C. urealyticum* and the genetically modified promotor P1 upstream to ioIT2 were analyzed for their ability to produce L-lysine from glucose by batch cultivation using the cultivation system according to Wouter Duetz.

[0145] As medium CGXII medium with 7.5 g/l CSL and containing 20 g/l glucose as carbon source was used. The cultures were incubated until complete consumption of glucose as confirmed by glucose analysis using blood glucose-meter. The cultivation time until complete sugar depletion, the concentration of L-lysine and the optical density OD660 were determined. The result of the experiment is presented in Table 8.

TABLE 8

L-lysine production and productivity.				
strain	OD660	L-lysine ¹ (g/l)	Time of sugar depletion (h) ²	Productivity (Lys g/l/h)
DM2042	5.4	8.8	18.6	0.47
DM2042_P1-ioIT	8.2	6.0	12.9	0.46
DM2042_pntAB_ec	5.1	9.3	19.9	0.47
DM2042_pntAB_ec_P1-ioIT	7.2	6.9	13.4	0.51
DM2042_pntAB_cu	4.7	9.9	21.1	0.47
DM2042_pntAB_cu_P1-ioIT	6.4	8.4	14.7	0.57

¹as L-lysine × HCl

²cultivation time until glucose = 0 g/l

Data represent mean values of four independent cultivations.

pntAB_ec and DM2042_pntAB_cu result in increased L-lysine titer (g/l) as compared to the parental strains DM2042.

[0147] Indeed, the overexpression of ioIT results in decreased L-lysine titer (g/l) in experiments with strains DM2042_P1-ioIT, DM2042_pntAB_ec_P1-ioIT and DM2042_pntAB_cu_P1-ioIT in comparison to the respective strains without ioIT overexpression but harboring the same pntAB variant.

[0148] Surprisingly, the combination of heterologous pntAB gene and overexpression of ioIT results in a strong increase of productivity.

Example 11

Comparison of the L-Lysine Production and Productivity Using Strains with Further Different Sugar Uptake Systems

[0149] Strains DM2042_pntAB_cu (reference) carrying the pntAB_cu genes derived from *C. urealyticum*, DM2042_pntAB_cu_P1-ioIT carrying the pntAB_cu genes derived from *C. urealyticum* and the genetically modified promotor P1 upstream to ioIT2, DM2042_pntAB_cu_P1-glf_zm carrying the pntAB_cu genes derived from *C. urealyticum* and the glf_zm gene derived from *Zymomonas mobilis*, DM2042_pntAB_cu_P2-ptsG carrying the pntAB_cu genes derived from *C. urealyticum* and the genetically modified promotor P2 upstream to ptsG, DM2042_pntAB_cu_P1-galP_ec carrying the pntAB_cu genes derived from *C. urealyticum* and the galP_ec gene derived from *E. coli* and strain DM2042_pntAB_cu_P1-bglF_cgR carrying the pntAB_cu genes derived from *C. urealyticum* and the bglF_cgR derived from *Corynebacterium glutamicum* were analyzed for their ability to produce L-lysine from glucose by batch cultivation using the cultivation system according to Wouter Duetz.

[0150] CGXII medium with 7.5 g/l CSL and containing 20 g/l glucose as carbon source was used for this experiment. The cultures were incubated until complete consumption of glucose as confirmed by glucose analysis using blood glucose-meter. The cultivation time until complete sugar depletion, the concentration of L-lysine and the optical density

[0146] The experiment shows that expression of heterologous pntAB_ec and pntAB_cu genes in strains DM2042_

OD660 were determined. The result of the experiment is presented in Table 9.

TABLE 9

L-lysine production and productivity.				
strain	OD660	L-lysine ¹ (g/l)	Time of sugar depletion (h) ²	Productivity (Lys g/l/h)
DM2042_pntAB_cu	4.7	9.9	21.1	0.47
DM2042_pntAB_cu_P1-iot	6.4	8.4	14.7	0.57
DM2042_pntAB_cu_P1-glf_zm	4.4	10.4	22.9	0.46
DM2042_pntAB_cu_P2-ptsG	6.7	6.2	28.1	0.22
DM2042_pntAB_cu_P1-galP_ec	5.0	9.6	20.1	0.48
DM2042_pntAB_cu_P1-bglF_cgR	4.8	10.1	21.4	0.47

¹as L-lysine x HCl²cultivation time until glucose = 0 g/l

Data represent mean values of four independent cultivations.

[0151] The experiment shows that only the overexpression of iot gene in strain DM2042_pntAB_cu_P1-iot results in a strong increase of productivity as compared to the parental strains DM2042_pntAB_cu. An increase in productivity has not been seen by analyzing the other strains comprising overexpression of genes glf, ptsG, galP and bglF, respectively.

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SEQ ID NO: 10 moltype = DNA length = 4839
 FEATURE Location/Qualifiers
 source 1..4839
 mol_type = other DNA
 organism = synthetic construct

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SEQ ID NO: 11 moltype = DNA length = 1527
 FEATURE Location/Qualifiers
 source 1..1527
 mol_type = genomic DNA
 organism = Corynebacterium glutamicum

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SEQ ID NO: 12 moltype = AA length = 508
 FEATURE Location/Qualifiers
 source 1..508
 mol_type = protein
 organism = Corynebacterium glutamicum

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SEQ ID NO: 13 moltype = DNA length = 1711
 FEATURE Location/Qualifiers
 source 1..1711
 mol_type = other DNA
 organism = synthetic construct

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atagttcgaa atcgggggca accccaatga aagaatcaga atcattgaac tagcgaggaa 600
atgggactca cacttcgtc cctccttctc ctaaaagcgt aggcgcggcca ccttcccgggt 660
tgttttaagt cgtttgaaaca agctttacaa gtgcacagggt ctaaatgctc caaacccctt 720
tcataaactc tgctttctgg atcgggttaag agctgcatat agcaacaatt gtgacttcat 780
caagataaaa atcgttaaca acacagacca aaacgggtcag ttaggtatgg atatcagcac 840
cttctgaacg ggtacgggta taatgggtggg cgtttgaaaa actcttcgcc ccacgaaaat 900
gaaggagcat aatgacggac atcaaggcca catcaagtac atcggccact acagcaccaa 960
cagcaggccg accagcgcgt cgacttggaac aaatttcctc cgtcgctgt ctggcgggac 1020
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gcacatggtt cgcgctcgtt gccctcctc cgtcgccct cttctcggc atgctgcgga 1560
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tgagagccgt cgtaccctc gagcgtgcga aagccgaaat ggatgaaatc atcgcggtgc 1680
actctgaaaa caatgcggca cttcctggtg t 1711

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SEQ ID NO: 14      moltype = DNA length = 2052
FEATURE
source             Location/Qualifiers
                   1..2052
                   mol_type = genomic DNA
                   organism = Corynebacterium glutamicum

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SEQUENCE: 14
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attgttgatc aacaagaaat tgactccgac ccatcagttc ttggcgtagt accccaagga 180
tccaccggta tgcagggtgt gatgggtgga tctgttgcaa actattacca agaaatcctc 240
aaacttgatg gaatgaagca ctgcgcgac ggtgaagcta cagagagttc atccaagaag 300
gaatacggcg gagtcgctgg caagtagctg tggattgact acgccttcga gttctgtct 360
gatactttcc gaccaatcct gtgggcccct cttggtgcct cactgattat tacctgtgtg 420
gttcttgctg attctttctg ttgcaagac ttccgcgctc caatggatga gcagcctgat 480
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ggtgccaccg cagctcgaaa gctcggcgca aacgagtgga ttggtgcagc tattccagcc 600
gcactttcta ctccagaatt ctgggcaact ggttctgccc gcgataccgt cacagtcttt 660
ggcctgccaa tggttctgaa tgactactcc ggacaggtat tcccaccgct gattgcagca 720
attgtctgtg actgggtgga aaaggagctg aagaagatca tccctgaagc agtccaaatg 780
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cgcaggcagc aagtcaggc gggggatcca ctgatcactt ttgacgtgta cttcattcga 1920
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aagaacgagt aa 2052

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SEQ ID NO: 15      moltype = AA length = 683
FEATURE
source             Location/Qualifiers
                   1..683
                   mol_type = protein
                   organism = Corynebacterium glutamicum

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SEQUENCE: 15
MASKLTTTSQ HILENLGGPD NITSMTHCAT RLRFQVKDQS IVDQQEIDSD PSVLGVVPQG 60
STGMQVVMGG SVANYQIEIL KLDGMKHPAD GEATESSSKK EYGGVRGKYS WIDYAFEFLS 120

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DTFRPILWAL	LGASLIITLL	VLADTFGLQD	FRAPMDEQPD	TYVFLHSMWR	SVFYFLPIMV	180
GATAARKLGA	NEWIGAAIPA	ALLTPEFLAL	GSAGDVTVTF	GLPMVLNDYS	GQVFPPLIAA	240
IGLYWVEKGL	KKIIPAVQM	VFPFFSLLI	MIPATAFLLG	PFQIGVNGI	SNLLEAINNF	300
SPFILSIVIP	LLYPFLVPLG	LHWPLNAIMI	QNINTLGYDF	IQGPMGAWNF	ACFGLVTGVF	360
LLSIKERNKA	MRQVSLGML	AGLLGGISEP	SLYGVLLRFK	KTYFRLLPGC	LAGGIVMGIF	420
DIKAYAFVFT	SLLTIPAMDP	WLGYTIGIAV	AFFVSMFLVL	ALDYRSNEER	DEARAKVAAD	480
KQAEEDLKAE	ANATPAAPVA	AAGAGAGAGA	GAAAGAATAV	AAKPKLAAGE	VVDIVSPLEG	540
KAIPLSEVPD	PIFAAGKLG	GIAIQPTGNT	VVAPADATVI	LVQKSGHAVA	LRLDSGVEIL	600
VHVGLDVTQL	GGEFTVHVE	RRQQVKAGDP	LITFDADFIR	SKDLPLITPV	VVSNAAKFGE	660
IEGIPADQAN	SSTTVIKVNG	KNE				683

SEQ ID NO: 16 moltype = DNA length = 1680
 FEATURE Location/Qualifiers
 source 1..1680
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 16

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ggaatggctg	actcctacga	cgagcagggt	aactacatct	tccctgaggg	cttcgacgcc	180
gagaccaacc	aatggctcga	aggcttcgat	gagcagcgtc	aggcttgagg	agctcgctac	240
gccgagtcgc	agcgtcgctt	caccgctcac	accgctcaga	tcgagcgctg	tcgtcagcag	300
gctgaagagg	cagctgcoga	ggctccggcc	ggcaactact	ccactgattc	tcgagaagat	360
gcacctgcag	cagaagcagt	tgaagagtcc	gctggctccc	tcgcttcoga	tgagcagctc	420
gctgctcttc	gcgagaagct	cgcaggtaac	taatagttcc	tgacacctct	aagagggtc	480
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aagtcgtgca	ggtcagggga	gtgttgcccg	aaaacattga	gaggaaaaca	aaaaccgatg	840
tttgattggg	ggaatcgtgt	ggtataatgg	taggacgcag	tgactgctat	cacccttggt	900
gggtctctgt	tgaaggaat	aattactcta	atggcggtcca	aactgacgac	gacatcgcaa	960
catattctcg	aaaaccttgg	tgaccagac	aatattactt	cgatgactca	ctgtgcgact	1020
cgcttcgct	tccaagtga	ggatcaatcc	attgttgatc	aacaagaaat	tgactccgac	1080
ccatcagttc	ttggcgtagt	agcccaagga	tccaccggta	tgacgggtgt	gatgggtgga	1140
tctgttgcaa	actattacac	agaaatcctc	aaacttgatg	gaatgaagca	cttcgccgac	1200
ggtagaagta	cagagagttc	atccaagaag	gaatacggcg	gagtcggtgg	caagtactcg	1260
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ggttctgcgc	gcgataccgt	cacagctctt	ggcctgccaa	tggttctgaa	tgactactcc	1620
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SEQ ID NO: 17 moltype = DNA length = 1422
 FEATURE Location/Qualifiers
 source 1..1422
 mol_type = genomic DNA
 organism = Zymomonas mobilis

SEQUENCE: 17

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ggcttgcttt	tcggttacga	ttcagcgggt	atcgctgcaa	tcggtacacc	ggttgatata	120
cattttattg	ccctcgtca	cctgtctgct	acggctgcgg	cttcccttcc	tgggtaggtc	180
gtttgtgctg	ttttggctcg	ttgtgttacc	ggttctttgc	tgtctggctg	gattggtatt	240
cgcttcggtc	gtcgcggcgg	attgttgatg	agttccattt	gtttcgtcgc	cgccggtttt	300
gggtctcgct	taaccgaaaa	attatttgga	acgggtggtt	cggctttaca	aattttttgc	360
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gatggttctc	cagcattgaa	tcagactttc	aaccacggtt	tctcctatct	cgttttcgca	1320
gcattaagta	cttaggtggg	cttgattgtt	gctcgtctcg	tgccggaac	caaaggtcgg	1380
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SEQ ID NO: 18 moltype = AA length = 473
 FEATURE Location/Qualifiers
 source 1..473
 mol_type = protein
 organism = *Zymomonas mobilis*

SEQUENCE: 18

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VVAVLVGCVT	GSLLSGWIGI	RFGRRGGLLM	SSICFVAAGF	GAALTEKLFQ	TGGSALQIFC	120
FFRFLAGLGI	GVVSTLTPTY	IAEIRPPDKR	QOMVSGQQMA	IVTGALTGYI	FTWLLAHFGS	180
IDWVNASGWC	WSPASEGLIG	IAFLLLLLTA	PDTPHWLVMK	GRHSEASKIL	ARLEPQADPN	240
LTIQKIKAGF	DKAMDKSSAG	LFAFGITVVF	AGVSVAAPQQ	LVGINAVLYY	APQMFQNLGF	300
GADTALLQTI	SIGVVFNFIT	MIASRVVDRF	GRKPLLIWGA	LGMAAMMAVL	GCCFWFKVGG	360
VLPLASVLLY	IAVFGMSWGP	VCWVVLSEMP	PSSIKGAAMP	IAVTGQWLAN	ILVNFLFKVA	420
DGSPALNQTF	NHGFSYLVFA	ALSILGGLIV	ARFVPETKGR	SLDEIEEMWR	SQK	473

SEQ ID NO: 19 moltype = DNA length = 3083
 FEATURE Location/Qualifiers
 source 1..3083
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 19

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cggtcatctt	ggagggtgca	aaagaaagcg	gtcatgccca	gctgatcgcg	gagccagcct	120
aatttttccc	gcaataaagt	tctgttaagc	agggcgctcaa	cggcactttc	gccaggtggg	180
aggatgttta	gcccaagccc	taacaagcca	ctgctgggtga	tgttgacgta	cccgacttcc	240
tctaacgaag	acgttattcg	atagatcgac	gacctcggct	catttaactg	ttttgccagc	300
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acccacaaca	caaaaagtct	tgacaaaagc	ttatcttacg	aacaagatgt	ccggtagatg	540
atagttcgaa	atcgggggca	accccaatga	aagaatcaga	atcattgaac	tagcgaggaa	600
atgggactca	caccttcgtc	cctccttctc	ctaaaagcgt	aggccggcca	ccttcccggg	660
tgttttaagt	cgtttgaaca	agctttacaa	gtgcacaggg	ctaaatgctc	caaacccttt	720
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caaaagataa	atcggttaaca	acacagacca	aaacgggtcag	ttaggtatgg	atatcagcac	840
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SEQ ID NO: 20 moltype = DNA length = 1395

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FEATURE
source

Location/Qualifiers
1..1395
mol_type = genomic DNA
organism = *Escherichia coli*

SEQUENCE: 20

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gcgggttata	ccaacactac	cgagcaaatg	tgggggacgg	tgattgtcgg	cctgaccaac	900
gtacttgcca	cctttatcgc	aatcggcctt	gttgaccgct	ggggacgtaa	accaacgcta	960
acgctgggct	tcctgggtgat	ggctgctggc	atgggcgtac	tcggtacaat	gatgcatatc	1020
ggtattcact	tccctgcggc	gcagtatttc	gccatcgcca	tgtgtctgat	gtttattgtc	1080
ggttttgcca	tgagtgcggg	tccgctgatt	tgggtactgt	gctccgaaat	tcagccgctg	1140
aaagcccgcg	attttgccat	caactgctcc	actgccacca	actggattgc	caacatgata	1200
gttgccgcaa	cgttcctgac	catgctcaac	acgctgggta	acgccaacac	cttctgggtg	1260
tatgcggctc	tgaacgtact	gtttatcctg	ctgacattgt	ggctggtagc	ggaacccaaa	1320
cacgtttcgc	tggaaacatat	tgaacgtaat	ctgatgaaa	gtcgtaaact	gcgcgaaata	1380
ggcgtcacg	attaa					1395

SEQ ID NO: 21

FEATURE
source

moltype = AA length = 464
Location/Qualifiers
1..464
mol_type = protein
organism = *Escherichia coli*

SEQUENCE: 21

MPDAKKQGRS	NKAMTFFVCF	LAALAGLLFG	LDIGVIAGAL	PFIADDEFQIT	SHTQEWVVS	60
MMFGAAVAV	SGSWLSFKLG	RKSLMIGAI	LPVAGSLFSA	AAPNVEVLIL	SRVLLGLAVG	120
VASYTAPLYL	SEIAPEKIRG	SMISMYQLMI	TIGILGAYLS	DTAFSYTGAW	RWMLGVIIIP	180
AILLLLGVFF	LPDSRWFVFA	KRRFVDAERV	LLRLRDTSAE	AKRELDEIRE	SLQVKQSGWA	240
LPKENSNNFR	AVFLGVLLQV	MQQFTGMNVI	MYAPKIFEL	AGYTNTTEQM	WGTIVIVGLTN	300
VLATPIAIGL	VDRWGRKPTL	TLGFLVMAAG	MGVLGTMHII	GIHSPSAQYF	AIAMLLMFIV	360
GFAMSAGPLI	WVLCSEIQPL	KGRDFGITCS	TATNWIANMI	VGATFLTMLN	TLGNANTFWV	420
YAALNVLPIL	LTLWLVPETK	HVSLEHIERN	LMKGRKLREI	GAHD		464

SEQ ID NO: 22

FEATURE
source

moltype = DNA length = 3056
Location/Qualifiers
1..3056
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 22

ccaggggcga	ggtaaatcag	atcaatatct	gcgcctcctc	gccaatccag	actgatgatg	60
cggatcatct	ggaggggtga	aaagaaagcg	gtcatgcccc	gctgatcgcg	gagccagcct	120
aatttttccc	gcaataaagt	tctgttaagc	agggcggtcaa	cggcactttc	gcccagggtg	180
aggatgttta	cgccaagccc	taacaagcca	ctgctggtga	tgttgacgta	cccgaacttc	240
tctaacgaag	acgttattcg	atagatcgac	gacctcggtc	catttaactg	ttttgccagc	300
tcagaatgag	tatttgcccc	tgccgaaaag	gcgtctacta	aaagtctctg	tttggtgagt	360
agctgaatgg	tttgcatatt	tgatccgttt	ttctaagggt	tgtttttgat	tttgcccccg	420
catgggggtg	tttgtgacgg	ctaacaaggt	tgtcttgggt	aactatatcg	acaaaaagta	480
acccacaaca	caaaaagtct	tgacaaaagc	ttatcttacc	aacaagatgt	ccggtagatg	540
atagttcgaa	atcgggggca	accccaatga	aagaatcaga	atcattgaac	tagcggagaa	600
atggggactca	caccttcgtc	cctccttctc	ctaaaaagcg	aggccgggcca	ccttcccggt	660
tgttttaagt	cgtttgaaaca	agctttacaa	gtgcacaggg	ctaaatgctc	caaacctctt	720
tccataactc	tgttttctgg	atcggttaag	agctgcatat	agcaacaatt	gtgacttcat	780
caaaagataaa	atcgtttaaca	acacagacca	aaacgggtcag	ttaggtatgg	atatcagcac	840
cttctgaacg	ggtagcgggt	taattggtgg	cgtttgaaaa	actcttcgcc	ccacgaaaat	900
gaaggagcat	aatgctgac	gctaaaaaac	agggcgcggtc	aaacaaggca	atgacgtttt	960
tcgtctgctt	ccttgccgct	ctggcgggat	tactctttgg	cctggatatc	ggtgtaattg	1020
ctggcgcact	gccgtttatt	gcagatgaat	tccagattac	ttcgcacacg	caagaatggg	1080
tcgtaagctc	catgatgttc	ggtagcggcg	tcgggtcggtg	gggcagcggc	tggctctcct	1140
ttaaactcgg	gcgcaaaaag	agcctgatga	tcggcgcaat	ttgtttgtgt	gccggttcgc	1200
tgttctctgc	ggctgcgcga	aacgttgaag	tactgattct	ttcccgcggt	ctactggggc	1260
tggcgggtgg	tgtggcctct	tataccgcac	cgctgtacct	ctctgaaatt	gcgcgggaaa	1320
aaattcgtgg	cagatgatgc	tcgatgtatc	agttgatgat	cactatcggtg	atcctcggtg	1380
cttatctctt	tgataccgcc	ttacgtatac	ccggtgcacg	gcgctggatg	ctgggtgtga	1440
ttatcatccc	ggcaattttg	ctgctgattg	gtgtcttctt	cctgccacag	agccacgctt	1500

-continued

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ggtttgccgc caaacgccgt tttgttgatg ccgaacgcgt gctgetacgc ctgcgtgaca 1560
ccagcgcgga agcgaaaacgc gaactggatg aaatccgtga aagtttgacg gttaaacaga 1620
gtggctgggc gctgtttaaa gagaacagca acttccgcgc cgcggtgttc cttggcgtag 1680
tgttgcaggt aatgcagcaa ttcaccggga tgaacgtcat catgtattac gcgccgaaaa 1740
tcttcgaact ggcggggttat accaacacta ccgagcaaat gtgggggacc gtgattgtcg 1800
gcctgaccac cgtacttgcc acctttatcg caatcgccct tgttgaccgc tggggacgta 1860
aaccaacgct aacgctgggc ttcctgggta tggctgctgg catggggcga ctcggtacaa 1920
tgatgcatac cggatttcac tctccgtcgg cgcagtattt cgcctcgcc atgctgctga 1980
tgtttattgt cggttttgcc atgagtgccg gtccgctgat ttgggtactg tgcctcgaaa 2040
ttcagccgct gaaaggccgc gattttggca tcacctgctc cactgccacc aactggattg 2100
ccaacatgat cgttggcgca acgttccctga ccatgctcaa cagctgggtt aacgccaaac 2160
ccttctgggt gtatgcccgt ctgaacgtac tgtttatcct gctgacattg tggctggtag 2220
cggaaccaa acacgtttcg ctggaacata ttgaacgtaa tctgatgaaa ggtcgtaaac 2280
tcgcgcaaat agcgcttcac gattaaaccc cctccgatct ctttggctag cacttaacaa 2340
accgccacac cttttagaaa gtgattctcg ctgtgaacaa ctactcgca ttaaacccag 2400
acaccctccc acagaaagtc atgtttggat atggcaagtc cagtgcattc ttaaagcagg 2460
aagtgaacgc ccgcggtcca gccaaagtcg tggctattgc gggtgaaacg gaaatgtcga 2520
tcgcccataa ggtggcctca gaaattgagg tggcgatctg gcacgcagaa gttgtcatgc 2580
acgtgcccat cgaagtagcc gaacgtgcgc gtgcagtggc aaccgacaat gagattgatc 2640
tgctgggtgtg ttttggcgac ggatccacca taggtttggc caaagcaat gccatgacca 2700
ctgccctgcc catcgtcgcg atccccacca cctacgcagg atcggaagca accaacgtgt 2760
ggggtctgac ggaagcagcg cgcaaaacaa ccggtgttga tctgaaggtg ctccccgaaa 2820
cagtcattta agattccgaa ctccacatgt cgcctccagt ggagatgtcc gtggcaccgc 2880
gactcaacgg cctggcgcac tgcattgatt ctttgtgggg acccaacgcc gatcccatca 2940
acgcagtgtc tgcagccgaa ggaatccgcg cactcaacca gggactgccg aaaattgttg 3000
cgaaacccga cagatcgaa ggacgcgacg aagccctcta cggcgctac ctgcga 3056

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SEQ ID NO: 23      moltype = DNA length = 1857
FEATURE            Location/Qualifiers
source              1..1857
                    mol_type = genomic DNA
                    organism = Corynebacterium glutamicum
                    strain = Corynebacterium glutamicum R

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SEQUENCE: 23
atggcaacga aaacagactt cagccccttg gtggctgatg ttgtcacgcg ttggggaggc 60
gctgacaacg tgcgcagtggt cagcgactgt gccaccaggc ttccgttcaa aatcaaggat 120
tccgacaagg ctgatatcgaa taaggctggc aaaatcagtg gcgtcattac tgcgatcaac 180
tctggtggac agcatcaggt tgtcatcgcc aatgatgttc cactggcgta tcaagcattg 240
atggatgtac ccggcatgcg cgcctaaaggc attaaagatg gcgatgcaga agtgagacat 300
gatggcggtt cagaggataa gaactggctc aacaagttca tcatctgat tctgcaactg 360
ttttcacgca tctgtggggc ttggcccggt atcggtttgg gtaaagctgg tcttaccctg 420
gcgaccactg taggaatttt ggatgcagaa tcaagcacc acaccatttt caacgcgatg 480
tttgatggct tgttttactt cttgccattc ttctcgcgca taacggctgc taagcgattt 540
aaagttaatc agttcattgc tctcgcaatt gtggccgcat tgcctgcacc cactatggaa 600
cgatgggtcg gtactgaggg agttacattc ctggggctac ctttgacct gatgtcttat 660
tcttcacag tgatccctgt ggttgtgtcg gtgtggttgg ccgataacct gcaacgctgg 720
ttggaaaagg tactgcttgg tctgttccgt aactcttca cgcactgat cgtgggtgtg 780
gtgatgggtc cgttgggtgt gtttaccatc ggcccagtc ccatcggtgc agcaaatgca 840
ctgtctaagt gccattgggt cctctttgaa acggttcctt ggttggccgg tggcctcatg 900
ggtgcattct gccaggtctt cgtgatgttt ggtctgcact ggggactggt gccagtgatg 960
atcaatgaca ttgcgaccca agccttctcc tacatgatgg cgcactgat ggctgcaatt 1020
ttcgacagcg tccgtgcggc tctagcagtg tggattagtg ccaagaacgc tgaagtgaac 1080
aaggtagctg gtccctgtgt cctttcagga tctctcgag gtgttactga accaatcatt 1140
tacggcgtga accttccact gaagtacccc ttctatgcgg gtgtcgtagc tgggtggaatt 1200
gggtgaatca tcatcgctgt ggggtggcaac gcttttgatg ccttcgtctt cccatcggtg 1260
atggctttcg cagcaacgct cagcgtgggt agcttcgtgg cgcaactaat tggttcgtca 1320
gctgcgctta tcttgggatt tgtgttcaat gtgggtgatg ttcgaaagga acaaacgctc 1380
gataagcaag aaaccgcac caccocaaat gtggctgcgg ccgcaagtgg cactacgtg 1440
ccgctggcag agattcttga caaggtgttt tcttcggag ccatgggcga tggatttggc 1500
atcgtgccgg aaaacggtga gattgtttca ccggtggaag gccagttgat tgcggtgcag 1560
aaatctgggc atgcatatgg catccgtggg gataatgga ttgaagttct ggtgacgtg 1620
ggcattgaca ccgtcaaaat ggcaggtgag ggattcgaaa tctcctgtggc gcgtggcgat 1680
cacgtcaaa gtgggcagcc gctggccaag gttgatttga atgtgttgc ggaggtggt 1740
ttcgatccga ccactgcggt gatcgtgact aactccaaga aactggaagg cttgaaattc 1800
actgatgttg agggacacgt tggcgagga gatgtagtga tgcgtatgga accttaa 1857

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SEQ ID NO: 24      moltype = AA length = 618
FEATURE            Location/Qualifiers
source              1..618
                    mol_type = protein
                    note = strain Corynebacterium glutamicum R
                    organism = Corynebacterium glutamicum

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SEQUENCE: 24
MATKTDSPSL VADVVTALGG ADNVRSVTHC ATRLRFKIKD SDKADIAKAG KISGVITAIN 60
SGGQHQVVG NDVPLAYQAL MDVPGMRAGK IKDGD AEVDD DGSSEDKNL NKFDLISAL 120
FSPIVWALAG IGLKGAGLTL ATTVGILDAE SSTHTIFNAM FDGLFYFLPF FLAITAAKRF 180

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KVNQFIALAI	VAALLHPTME	AMVGTGTVTF	LGLPLTMSY	SSSVIPVVVS	VWLAGYLQRW	240
LEKVLPGAAR	NFFTPLIVVL	VMVPLVLFIT	GPVTIGAANA	LSNGIGFLFE	TVPWLAGGLM	300
GAFWQVVFVF	GLHWGLVPVM	INDIATQGF	YMMAPLMAAI	FGQVGAALAV	WIRSKNAEVK	360
KVAGPGVLSG	FFAGVTEPII	YGVNLPLKYP	FYAGVVAGGI	GGIIIAVGGN	AFDAFVFPVS	420
MAFAATLSVG	SFVAQLIGSS	AALILGFVFT	VVMFRKEQNV	DKQETDTPN	VAAAAAGTIV	480
PLAEIPDKVF	SSGAMGGDGF	IVPENGEIVS	PVEGQLIAVQ	KSGHAYGIRG	DNGIEVLVHV	540
GIDTVKMAGE	GFELLVARGD	HVKVGQPLAK	VDLNAVAEAG	PDPTTAVIVT	NSKKLEGLKF	600
TDVEGHVGAG	DVVMRMEP					618

SEQ ID NO: 25 moltype = DNA length = 3518
 FEATURE Location/Qualifiers
 source 1..3518
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 25

ccagggcgga	ggtaaatcag	atcaatatct	gcgcctcctc	gccaatccag	actgatgatg	60
cggatcatct	ggaggggtgca	aaagaaagcg	gtcatgcccc	gctgatcgcg	gagccagcct	120
aatttttccc	gcaataaagt	tctgttaagc	agggcggtcaa	cggcactttc	gcccaggttg	180
aggatgttta	cgccaagccc	taacaagcca	ctgctggtga	tggtgacgta	cccgaacttc	240
tctaaccgaag	acgttatctg	atagatcgac	gacctcggtc	catttaactg	ttttgcagcg	300
tcagaatgag	tatttggccc	tgccgaaagt	gcgtctacta	aaagtctctg	tttggtgagt	360
agctgaatgg	tttgcattat	tgatccgttt	ttctaaaggt	tggttttgat	ttgcccccg	420
catggggtgt	tttgtgacgg	ctaaacaggt	tgtcttggtg	aactatatcg	acaaaaagta	480
accacacaac	caaaaagtct	tgacaaaagc	ttatcttacg	aacaagatgt	ccggtagatg	540
atagttcgaa	atcgggggca	accocaaatga	aagaatcaga	atcattgaac	tagcgaaggaa	600
atgggactca	caccttcctc	cctcctcttc	ctaaaagcgt	aggccggcca	ccttcccggg	660
tggttttaagt	cggttgaaac	agctttacaa	gtgcacaggg	ctaaatgctc	caaacctctt	720
tcataaactc	tgctttctgg	atcgggttaag	agctgcatac	agcaacaatt	gtgacttcat	780
caaagataaa	atcgtttaaca	acacagacca	aaacgggtcag	ttaggtatgg	atatcagcac	840
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ctcgatcaaa	ctctggtgga	cagcatcagg	ttgtcatcgg	caatgatgtt	ccactggcgt	1140
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aagtggacga	tgatggcggt	tcagaggata	agaactggct	caacaagtct	atcgatctga	1260
tttctgcact	gttttcaacc	atcgtgtggg	ctttggccgg	tatcgggttg	ggtaaaagctg	1320
gtcttaccct	ggcgaccact	gtaggaaatt	ttggatgcaga	atcaagcacc	cacaccattt	1380
tcaacgcgat	gtttgatggc	ttgttttact	ctctgccatt	cttcctcgcg	ataacggctg	1440
ctaagcgatt	ttaagttaat	cagttcattg	ctctcgcaat	tggtggccga	ttgctgcacc	1500
ccacgatgga	agcgtgggtc	ggtagtgagg	gagttacctt	cctggggcta	cctttgacca	1560
tgatgtctta	ttcttcacga	gtgatccctg	tggtgtgtgc	gggtgtggtg	gcccagatac	1620
tgcaacgcgtg	tttggaaaaag	gtactgcctg	gtgctgtccg	taactttctc	acgccactga	1680
tcgtgggtgt	ggtagatggt	ccgttggtgc	tggtttaccat	cgccccagtc	accatcggtg	1740
cagcaaatgc	actgtctaat	ggcattgggt	tcctctttga	aacgggttct	tggttgcccg	1800
gtggcctcat	gggtgcattc	ttgcaggtct	tcgtgatgtt	tggtctgcac	tggggactgg	1860
tgccagtgat	gatcaatgac	attgcgaccc	aaggttcttc	ctacatgatg	gcgccactga	1920
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ctggtggaat	tggtggaatc	atcatcgctg	tggtgggcaa	cgcttttgat	gccttcgtct	2160
tcctatcggt	gatggctttc	gcagcaacgc	tcagcgtggg	tagcttcgtg	gcgcaactaa	2220
ttggttcgtc	agctgcgctt	atcttgggat	ttgtgttcac	tggtgtgatg	tttcgaaaagg	2280
aacaaaacgt	cgataagcaa	gaaaccgaca	ccaccccaaa	tggtggtcgc	gcccgaagtg	2340
gcacctacgt	ccgctgggca	gagattcctg	acaaggtgtt	ttctcccgga	gccatgggcg	2400
atggatttgg	catcgtgcgc	gaaaacgggtg	agattgtttc	accggtggga	ggccagttga	2460
ttgcggtgca	gaaatctggg	catgcatatg	gcattccgtg	ggataatgga	attgaagttc	2520
tggtgcacgt	ggcatttgac	accgtcaaaa	tgccaggtga	gggattcgaa	atcctcggtg	2580
cgcgtggcga	tcacgtcaaa	gttgggcagc	cgcgtggcaa	gggtgatttg	aatgctgttg	2640
cggagggctgg	tttcgatccg	accactgcgg	tgatcgtgac	taactccaag	aaactggaag	2700
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cagcaagggt	catggtcatt	gcgggtgaac	gagaaatgtc	gatcgcccat	aagggtggct	3000
cagaaattga	ggtggcgatc	tgccacgacg	aagttgtcat	gcacgtgccc	atcgaagtag	3060
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cgatccccac	cacctacgca	ggatcggaag	caaccaacgt	gtggggctcg	acggaagcag	3240
cgcgcaaaac	aaccggtgtt	gatctgaagg	tgctccccga	aacagtcatt	tacgattccg	3300
aaactcacat	gtcgtctcca	gtggagatgt	ccgtggcacc	cggactcaac	ggcctggcgc	3360
actgcattga	ttctttgtgg	ggacccaacg	ccgatcccat	caacgcagtg	cttgacggcg	3420
aaggaaatccg	cgcactcaac	cagggactgc	cgaaaattgt	tgccgaacccg	cacagcatcg	3480
aaggacgcga	cgaagccctc	tacggcgctc	acctcgca			3518

SEQ ID NO: 26 moltype = DNA length = 1266

-continued

FEATURE
source
Location/Qualifiers
1..1266
mol_type = genomic DNA
note = Exchange of nucleobase cytosine (c) at position 932 of lysC gene sequence derived from ATCC13032 wildtype to give thymine (t). The acc codon for threonine is altered to the atc codon for isoleucine.
organism = Corynebacterium glutamicum

SEQUENCE: 26

gtggccctgg	tcgtacagaa	atatggcggt	tcctcgcttg	agagtgcgga	acgcattaga	60
aacgtcgctg	aacggatcgt	tgccaccaag	aaggctggaa	atgatgcgt	ggttgtctgc	120
tccgcaatgg	gagacaccac	ggatgaactt	ctagaacttg	cagcggcagt	gaatcccgtt	180
ccgccagctc	gtgaaatgga	tatgctctct	actgctggtg	agcgtatttc	taacgctctc	240
gtcgccatgg	ctattgagtc	ccttggcgca	gaagcccaat	ctttcacggg	ctctcaggct	300
ggtgtgctca	ccaccgagcg	ccaccgaaac	gcacgcattg	ttgatgtcac	tccaggctcg	360
gtgcgtgaag	cactcgatga	gggcaagatc	tgcattgttg	ctggtttcca	gggtgttaat	420
aaagaaaccc	gcgatgtcac	cacgttgggt	cgtggtggtt	ctgacaccac	tgcagttgcg	480
ttggcagctg	ctttgaacgc	tgatgtgtgt	gagatttact	cggacgttga	cggtgtgtat	540
accgctgacc	cgcgcacgtg	tcctaatagc	cagaagctgg	aaaagctcag	cttcgaagaa	600
atgctggaa	ctgtctgctg	tggctccaag	attttgggtg	tgcgcagtgt	tgaatacgct	660
cgtgcattca	atgtgccact	tcgcgtacgc	tcgtcttata	gtaatgatcc	cggcactttg	720
attgccggct	catggagga	tatcctgtg	gaagaagcag	tcctaccgg	tgtcgcaacc	780
gacaagtccg	aagccaaagt	aaccgttctg	ggtatttccg	ataagccagg	cgaggctcgc	840
aaggttttcc	gtgcgttggc	tgatgcagaa	atcaacattg	acatggttct	gcagaacgtc	900
ttctctgtag	aagacggcac	caccgacatc	atcttcacct	gccctcgttc	cgacggccgc	960
cgcgcgatgg	agatcttgaa	gaagcttcag	gttcagggca	actggaccaa	tgtgctttac	1020
gacgaccagg	tcggcaaaagt	ctccctcgctg	ggtgctggca	tgaagtctca	cccagggtgt	1080
accgcagagt	tcattggaagc	ttgcgcgat	gtcaacgtga	acatcgaatt	gatttccacc	1140
tctgagattc	gtatttccgt	gctgatccgt	gaagatgac	tggatgctgc	tgcacgtgca	1200
ttgcatgagc	agttccagct	gggcggcgaa	gacgaagccg	tcgtttatgc	aggcaccgga	1260
cgctaa						1266

SEQ ID NO: 27
FEATURE
source
moltype = AA length = 421
Location/Qualifiers
1..421
mol_type = protein
organism = Corynebacterium glutamicum

SEQUENCE: 27

MALVVQKYGG	SSLESAERIR	NVAERIVATK	KAGNDVVVVC	SAMGDTTDEL	LELAAAVNPV	60
PPAREMDMLL	TAGERISNAL	VAMAIESLGA	EAQSFTGSQA	GVLTERHGN	ARIVDVTGPR	120
VREALDEGKI	CIVAGFQGVN	KETRDVTTLG	RGGSDTTAVA	LAAALNADVC	EIYSDVDGVY	180
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Location/Qualifiers
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organism = Corynebacterium glutamicum

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VLIEAGFSEN AALIANVAPG VIAVVGAFIA LWMMDRINRR TTLITGYSLT TISHVLIGIA 360
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1. A *C. glutamicum* strain having the ability to produce L-lysine, the *C. glutamicum* strain comprising:

- a heterologous gene encoding a NAD(P)(+) transhydrogenase subunit alpha,
- a heterologous gene encoding a NAD(P)(+) transhydrogenase subunit beta, and
- a gene encoding a myo-inositol permease that is overexpressed by increasing a copy number of the gene, functionally linking the gene with a strong promoter, enhancing a ribosomal binding site, by codon usage optimization of a start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of a repressor binding region of the strong promoter or by a combination of two or more of these measures.

2. The *C. glutamicum* strain of claim 1, wherein the ability to produce L-lysine is achieved by overexpression of genes coding for enzymes having a function of an aspartate-semialdehyde dehydrogenase, an aspartate aminotransferase, a dihydrodipicolinate synthase, a dihydrodipicolinate reductase, a diaminopimelate decarboxylase, an aspartokinase and of a diaminopimelate dehydrogenase, wherein the overexpression is achieved by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the strong promoter or by a combination of two or more of these measures.

3. The *C. glutamicum* strain of claim 2, wherein the aspartokinase is feedback-resistant.

4. The *C. glutamicum* strain of claim 2, further comprising:

a gene coding for a pyruvate carboxylase that is overexpressed by increasing the copy number of the gene, functionally linking the gene with a strong promoter, enhancing the ribosomal binding site, by codon usage optimization of the start codon or of up to the whole gene, inactivation or deletion of a repressor gene, mutation of the repressor binding region of the strong promoter or by a combination of two or more of these measures.

5. The *C. glutamicum* strain of claim 1, wherein an amino acid sequence of the NAD(P)(+) transhydrogenase subunit alpha has an identity of at least 60% to an amino acid sequence according to SEQ ID NO: 7.

6. The *C. glutamicum* strain of claim 1, wherein an amino acid sequence of the NAD(P)(+) transhydrogenase subunit beta has an identity of at least 70% to an amino acid sequence according to SEQ ID NO: 8.

7. The *C. glutamicum* strain of claim 1, wherein an amino acid sequence of the myo-inositol permease has an identity of at least 57% to an amino acid sequence according to SEQ ID NO: 12.

8. A method for the fermentative production of an L-lysine, the method comprising:

cultivating the *C. glutamicum* strain of claim 1, and accumulating L-lysine in a medium to form an L-lysine containing fermentation broth.

9. The method of claim 8, further comprising: manufacturing an L-lysine containing product from said L-lysine containing fermentation broth.

10. The method of claim 8, further comprising: isolating L-lysine from the L-lysine containing fermentation broth.

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